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| 2 | <p>Fig. 2.1 shows a small conducting sphere suspended from a long insulating thread between two metal plates M and M' that are 0.0500 m apart. The plates are connected to a 5.0×10^3 V battery. The sphere has a radius of 0.0025 m and a mass of 1.0 g.</p> |
|----------|---|

[Turn over

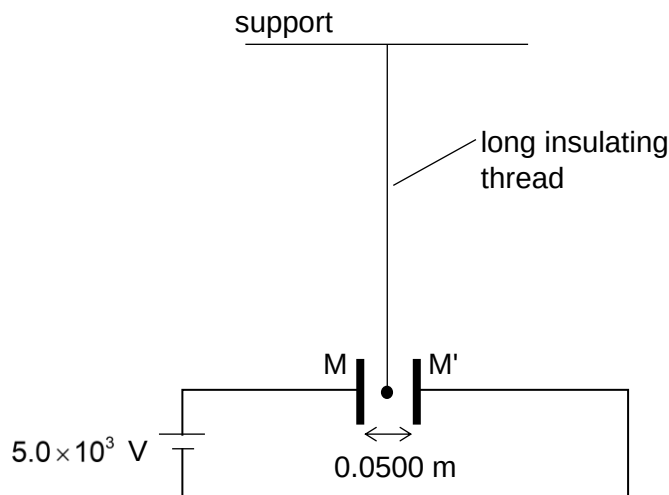


Fig. 2.1

The sphere is given an initial displacement such that the sphere touches M. It then moves rapidly to M', touches it, and returns rapidly to M again. This process repeats itself.

- (a) State and explain why an initial displacement to touch one of the plates is necessary for the process to start.

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[2]

When sphere is at the centre, the attractive force between sphere and M and Sphere and M' is equal and opposite.
There is translational equilibrium. The sphere does not have a resultant acceleration to move to M or M' [B1]

A displacement to M will result in sphere being positively charged, repelling M and attracted toward M' thus resulting in an acceleration towards M. This velocity will keep increasing as sphere move towards M. Constant electric Force/Field = constant acceleration [B1]

Comments:

Most students did not explain why without displacement, the oscillation cannot occur, and why with touching one of the plates, the oscillations occurs.

- (b) When the sphere touches either plate, it acquires a potential that is equal to the potential difference between the plates.

The electrical potential on the surface of a charged conducting sphere can be determined by assuming that all its charges are accumulated at the centre of the sphere.

Show that the charge on the sphere when it touches M is $50\pi\epsilon_0$.

[1]

		$V = \frac{Q}{4\pi\epsilon_0 r}$ $5.0 \times 10^3 = \frac{Q}{4\pi\epsilon_0 (0.0025)}$ $Q = 50\pi\epsilon_0 \text{ (proved)}$ Comments: Most students are able to prove this part.			
	(c)	If the electric field between the plates is uniform, calculate the magnitude of the electrostatic force acting on the sphere.			
			force =	N	[1]
		$F = QE = Q \frac{V}{d} = (50\pi \times 8.85 \times 10^{-12}) \frac{5.0 \times 10^3}{0.0500} = 1.39 \times 10^{-4} \text{ N}$ Comments: Most students are able to answer this part.			
	(d)	As a long thread is used, the motion of the sphere is nearly horizontal and is due to electrostatic force only. Determine the time taken for the sphere to move from M to M'.			
			time taken =	s	[3]

